Nihad EL HAMRY

Data Engineering & Artificial Intelligence Engineering Student

Seeking a 6-Month End-of-Studies Internship (PFE) Starting January 2027

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Portfolio | LinkedIn | GitHub

PROFILE

Final-year engineering student (5th year) at ENSA Fez in Data Science & Artificial Intelligence, seeking a 6-month End-of-Studies Internship starting January 2027 in Data Engineering and/or AI. My projects cover data pipeline design, Business Intelligence, Machine Learning, and Generative AI.

EDUCATION

2024 – Present — **Engineering Degree in Data Science & AI**
National School of Applied Sciences (ENSA) — *Fez, Morocco*

2022 – 2024 — **Integrated Preparatory Cycle**
National School of Applied Sciences (ENSA) — *Fez, Morocco*

2022 — **High School Diploma in Physical Sciences**
Lycée Moulay Rachid — *Fez, Morocco*

PROFESSIONAL EXPERIENCE

Data Engineering, BI & AI Intern — **July – August 2026**
Acacia Afrique Système (acacia-afrique.com) — *Morocco*

- **Data Engineering & BI:** exploration of Sage X3 data on SQL Server, development of ETL pipelines using Apache Hop, and creation of Power BI reports.
- **Automation:** scheduling Apache Hop workflows and configuring Power BI Gateway to automate data refresh in Microsoft Fabric.
- **Generative AI & Multi-Agent Systems:** development of a virtual enterprise simulation with CrewAI, integration of models via Ollama, and implementation of a RAG approach to leverage corporate documents.

AI & NLP Developer Intern — **July – August 2025**
AriMayi (arimayi.fr) — *France*

- Development of the core logic of a multilingual chatbot (French, English, and Arabic) using Rasa and FastAPI, featuring a fallback LLM (Llama 3 via Ollama).
- Management of data persistence and knowledge storage using SQLite.
- Integration of the solution into the existing platform and version control management with GitHub.

PROJECTS

Olist E-commerce Platform — ***Data Engineering & AI***

- **Cloud & Ingestion:** loading Kaggle datasets onto Amazon S3, configuring a S3–Snowflake storage integration, and ingesting raw data into RAW.
- **Transformation & Modeling:** exploratory analysis on Snowflake, dbt transformations in STAGING and MARTS, and implementation of snapshots.
- **AI & Orchestration:** development of a Python script to classify reviews, detect dissatisfaction, and identify root causes from comments; orchestration with Apache Airflow under Docker and creation of Power BI dashboards.

RealTime-Aero-Maintenance — ***Data Engineering & Machine Learning***

- Streaming ingestion simulation with Apache Kafka on the NASA C-MAPSS FD001 dataset, transmission of data to the Remaining Useful Life (RUL) prediction model, and display of predictions and alerts on a dashboard.
- Development and evaluation of CatBoost/SVR models, achieving a MAE of 10.25 cycles according to the evaluation protocol used; storage with Hadoop HDFS and containerization with Docker.

Enterprise Knowledge Brain — ***RAG & NLP***

- Development of a RAG assistant based on Llama 3, text extraction with Tesseract OCR, and semantic indexing in ChromaDB.
- Workflow orchestration with LangChain and creation of a user interface using Streamlit.

Decision Support System (SNCF Project) — ***Data Engineering & BI***

- Design of a star-schema Data Warehouse and development of ETL flows with Talend Open Studio.
- Multidimensional analysis using OLAP cubes under SSAS and creation of Power BI dashboards.

TECHNICAL SKILLS

- **Data Engineering & Big Data:** SQL, Snowflake, dbt, Apache Hop, Talend, Apache Kafka, Hadoop HDFS, Apache Airflow, AWS S3.
- **AI, ML & NLP:** Machine Learning, RAG, NLP, LangChain, CrewAI, Rasa, Ollama, CatBoost, scikit-learn, Tesseract OCR.
- **Languages & Databases:** Python, Pandas, NumPy, SQL, SQLite, SQL Server, MySQL.
- **BI, Development & Tools:** Power BI, Microsoft Fabric, Tableau, FastAPI, Streamlit, Docker, Git/GitHub, Chart.js.

CERTIFICATIONS

- **Cloud:** AWS Cloud Practitioner Essentials (in progress).
- **AI:** LangChain – Developing LLM Applications (DataCamp, 2026).
- **Machine Learning:** Machine Learning with Python (IBM/Cognitive Class, 2026).
- **Databases:** SQL Programming & Database Design (Oracle, 2024).

LANGUAGES & SOFT SKILLS

Languages
Arabic: native | French: fluent
English: professional / technical

Soft Skills
Problem solving | Analytical thinking
Teamwork | Technical curiosity